Hierarchical BERT-Based Multi-Label Models for Mashup API Recommendation

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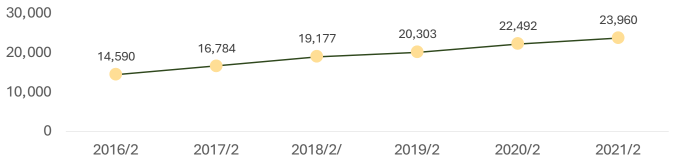
Introduction

- In recent years, the number of available Web services has rapidly increased [1].
- A service created by combining multiple Web services is called a **Mashup service**.



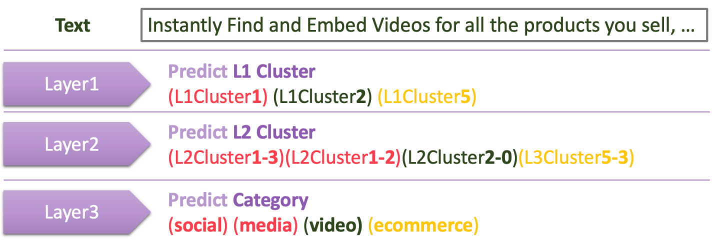
Source: J. J. Omena, 2019

- Necessary to select appropriate services from a huge number of candidates
- To enable easy mashup construction → ***Service composition***
- Discover appropriate Web services from natural language requirements

Method

- Propose combinations of services from requirements using BERT
- Formulate mashup API recommendation as multi-label classification

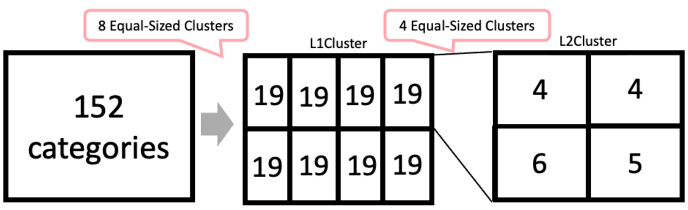
- However, accuracy is low due to the very large number of services to classify
- Accuracy also decreases due to label imbalance
- Solve this by hierarchical multi-label classification



- **Hierarchical BERT-Based Multi-Label Models**

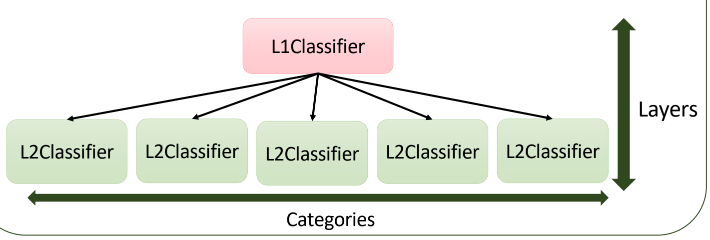
1：Hierarchical clustering

- Vectorize category names and cluster step by step



2：Layers and categories

- The number of layers and categories are hyperparameters
- **Layers**: Number of models to be traversed until the final category is predicted
- **Categories**: Number of categories distinguished by one model at each layer



Evaluation

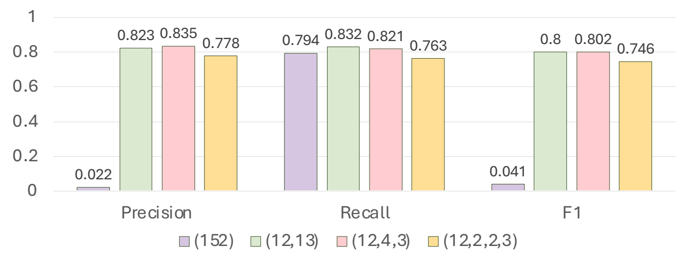
Data obtained from ProgrammableWeb [1]
Training data: 19,746 samples Validation data: 635 samples

of categories in each layer	Precision	Recall	F1
(152) Baseline	0.022	0.794	0.041
(6,25)	0.677	0.676	0.632
(12,13)	0.823	0.832	0.8
(15,10)	0.663	0.687	0.629
(24,7)	0.652	0.688	0.629
(33,4)	0.609	0.714	0.609
(51,3)	0.642	0.698	0.619
(4,4,8)	0.702	0.651	0.64
(8,4,5)	0.702	0.687	0.658
(12,2,6)	0.787	0.77	0.754
(12,4,3)	0.835	0.821	0.802
(12,6,2)	0.79	0.769	0.755
(16,4,2)	0.693	0.682	0.644
(12,2,2,3)	0.778	0.763	0.746

Baseline: Single-layer model directly outputting 152 categories
Accuracy improved by hierarchical multi-label classification
Especially (12, 4, 3) shows the highest accuracy

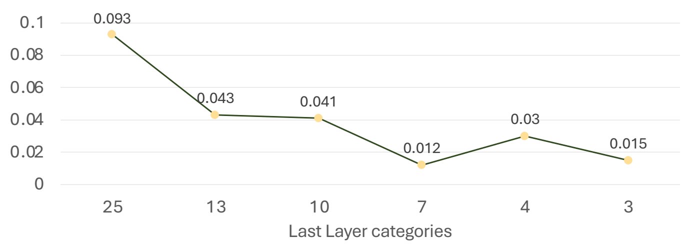
Analysis

- **1 : Number of layers**
- Compared the highest F1 scores across Number of layers



- More layers → fewer categories per layer, easier classification.
- But more layers → accumulated misclassification risk, lower accuracy.
- 2 Layers → less accumulated risk but harder classification → low accuracy
- 4 Layers → easier classification but high accumulated risk → low accuracy
- Among the tested patterns, (12,4,3) showed the best balance.

- **2 : Number of categories**
- Impact of the number of categories in the last layer
- Compare F1 score when only L1 category is predicted by L1 model with the F1 score when final categories are predicted, per number of categories



- More final-layer categories → lower L1→L2 accuracy
- Fewer final-layer categories → higher L1→L2 accuracy
- Reducing final-layer categories preserves accuracy

- **Future Work**
- Optimization of layers and categories using GA
- Service recommendation using category labels as input

- **References**

[1] Janna Joceli Omena, “Web APIs over the years,” blog post, Mar. 22, 2019. [Online]. Available: https://thesocialplatforms.wordpress.com/2019/03/22/web-apis-over-the-years/

1. http://www.programmableweb.com/